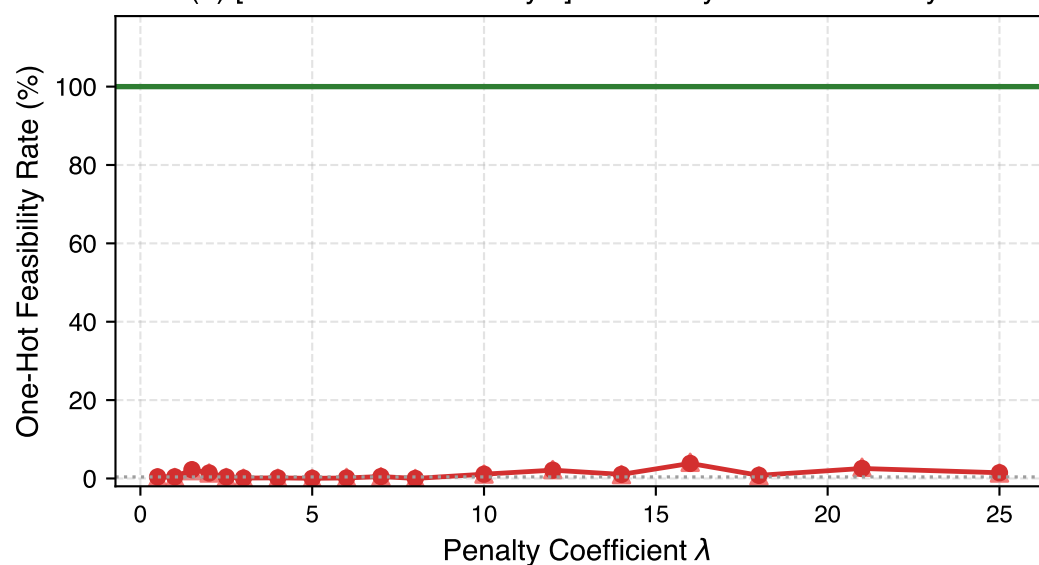
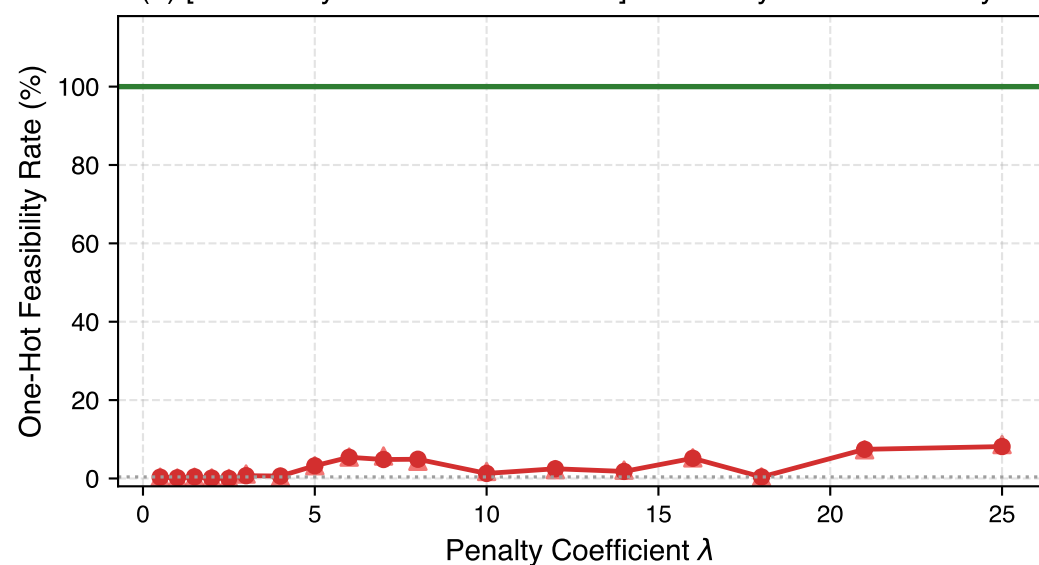
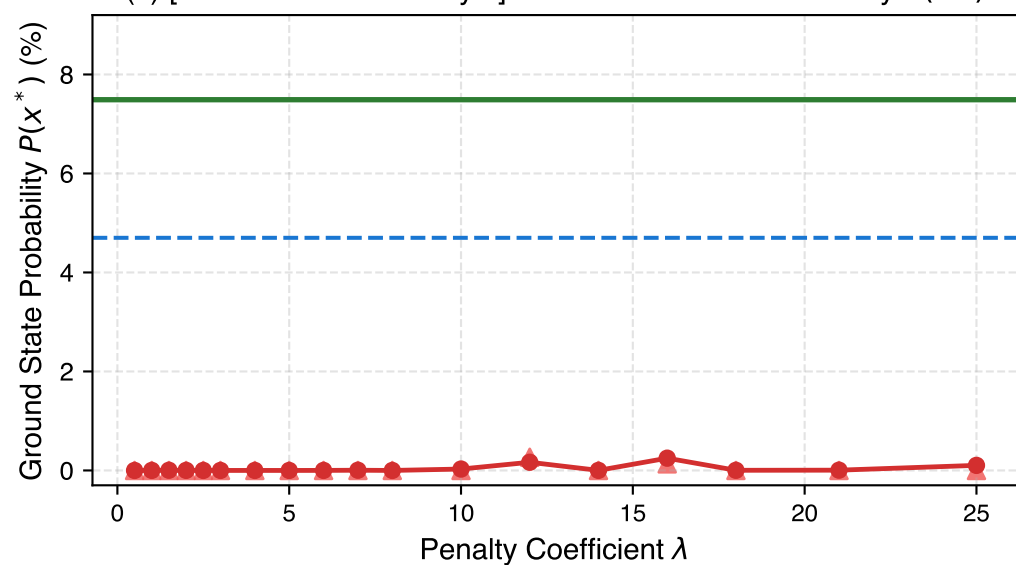


(a) [BB-1 Materials Catalyst] Feasibility Rate vs Penalty λ 

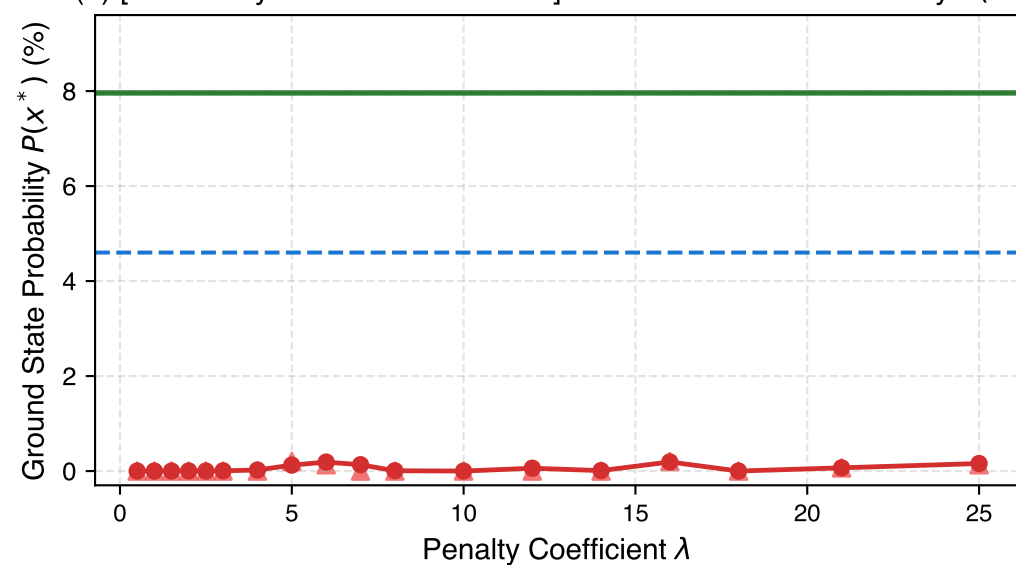
● Std QAOA (Theoretical State Vector) — FM-XY-QAOA [Proposed] (100% Inherent, $\lambda = 0$)
 ▲ Std QAOA (1500 Shots Sampled) Uniform Random Baseline ($256/2^{16} = 0.39\%$)

(b) [BB-2 Fully-Connected Random] Feasibility Rate vs Penalty λ 

● Std QAOA (Theoretical State Vector) — FM-XY-QAOA [Proposed] (100% Inherent, $\lambda = 0$)
 ▲ Std QAOA (1500 Shots Sampled) Uniform Random Baseline ($256/2^{16} = 0.39\%$)

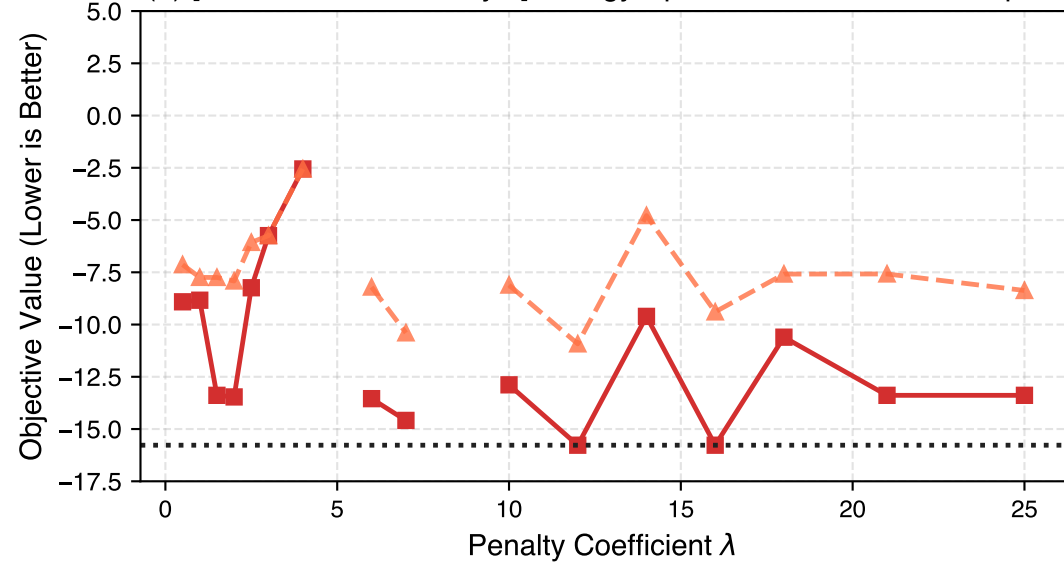
(c) [BB-1 Materials Catalyst] Ground State Hit Probability $P(x^*)$ 

● Std QAOA $P(x^*)$ (Theoretical) — FM-XY-QAOA [Proposed] (7.49%, $\lambda = 0$)
 ▲ Std QAOA (Sampled Hits) - - - FMQA (SA, Adaptive $\lambda = 8.17$, 4.70%)

(d) [BB-2 Fully-Connected Random] Ground State Hit Probability $P(x^*)$ 

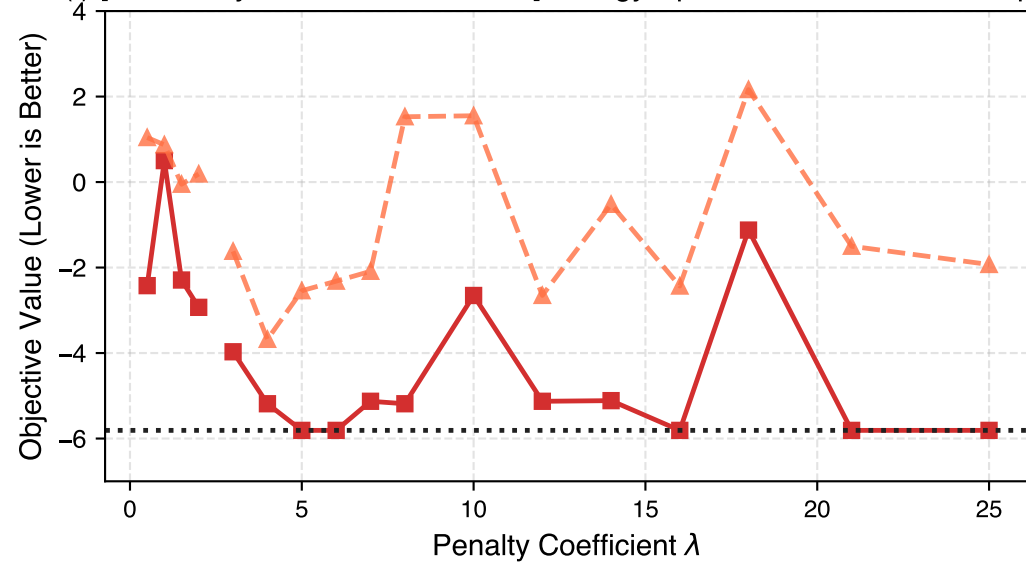
● Std QAOA $P(x^*)$ (Theoretical) — FM-XY-QAOA [Proposed] (7.96%, $\lambda = 0$)
 ▲ Std QAOA (Sampled Hits) - - - FMQA (SA, Adaptive $\lambda = 3.15$, 4.60%)

(e) [BB-1 Materials Catalyst] Energy Spectrum of Feasible Samples



■ Std QAOA Best Feasible Found ▲ Std QAOA Mean Feasible Energy
 Global Optimum $f^* = -15.77$

(f) [BB-2 Fully-Connected Random] Energy Spectrum of Feasible Samples



■ Std QAOA Best Feasible Found ▲ Std QAOA Mean Feasible Energy
 Global Optimum $f^* = -5.81$